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A Comparative Study of Unsupervised Machine Learning and Deep Learning Techniques for Anomaly Detection in Recommender Systems

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Abstract

Recommender systems are increasingly exposed to anomalous user behavior that can distort recommendation outcomes and compromise system reliability. In real-world settings, explicit labels identifying malicious activity are rarely available, motivating the adoption of unsupervised detection approaches. This study presents a systematic comparative analysis of classical machine learning and deep learning techniques for anomaly detection in recommender systems. Using the MovieLens 1M dataset, we construct a user-level behavioral representation based on statistical, temporal, and interaction-based features derived from explicit rating data. Three unsupervised detection models are evaluated: Isolation Forest, One-Class Support Vector Machine, and an autoencoder-based neural network. To address the absence of ground-truth labels, evaluation is conducted using a comprehensive label-free protocol, including score distribution analysis, percentile-based thresholding, ranking stability, and inter-model agreement. In addition, controlled experiments with synthetic attack profiles are conducted to assess detection performance under different adversarial strategies. Results indicate that individual models capture complementary aspects of anomalous behavior, exhibiting low to moderate agreement. An ensemble scoring strategy improves ranking stability and provides a consistent mechanism for identifying highly deviant user profiles. The findings suggest that ensemble-based unsupervised detection constitutes a practical and interpretable first-layer screening approach for recommender system monitoring under label-scarce conditions.

Keywords: recommender systems; anomaly detection; unsupervised learning; ensemble methods; user behavior analysis



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1. Introduction

Recommender systems have become a foundational component of modern digital ecosystems, shaping user experiences across e-commerce platforms, streaming services, social networks, and online marketplaces [1]. By leveraging historical user-item interaction data, these systems aim to provide personalized recommendations that enhance engagement, satisfaction, and platform retention. As recommender systems increasingly influence user decisions and market visibility, their robustness and trustworthiness have become critical concerns.

However, the widespread adoption of recommender systems has also exposed them to adversarial manipulation. One of the most extensively studied threats is the presence of malicious or strategic users who attempt to influence recommendation outcomes by injecting biased interactions, commonly referred to as shilling or profile injection attacks [2,3]. Such attacks can distort ranking algorithms, unfairly promote specific items, and degrade the reliability of collaborative filtering models. Early empirical studies demonstrated that even a small fraction of malicious profiles can significantly alter recommendation outputs, particularly in sparse data environments [2,4]. These findings established the vulnerability of collaborative recommender systems and motivated the development of defensive mechanisms.

Traditional countermeasures against shilling attacks have primarily relied on rule-based heuristics or supervised classification approaches that assume the availability of labeled attack data [4]. While these methods can be effective in controlled experimental settings, they face substantial limitations in real-world deployments. Explicit annotations identifying malicious users are rarely available, attack strategies evolve over time, and adversarial behavior often overlaps with legitimate but atypical user activity [3]. These challenges highlight the difficulty of maintaining static defense mechanisms in dynamic recommendation environments.

In response, anomaly detection has emerged as a promising paradigm for recommender system security [5,6]. Rather than modeling predefined attack classes, anomaly detection approaches conceptualize malicious behavior as deviations from typical interaction patterns. Tree-based, kernel-based, and neural network-based unsupervised models have demonstrated the ability to capture irregular rating distributions, abnormal temporal activity, and unusual interaction patterns without requiring labeled training data [5]. Nevertheless, much of the existing literature evaluates detection techniques in isolation or under synthetic attack injection scenarios, limiting understanding of how heterogeneous detection paradigms compare and interact under consistent feature representations.

Despite the availability of numerous anomaly detection techniques, there is a lack of standardized evaluation frameworks that enable consistent comparison across models under realistic constraints, such as the absence of labeled attack data. In practice, recommender systems must operate under label-scarce conditions, where both the nature and prevalence of adversarial behavior are uncertain. This gap motivates the need for systematic and reproducible evaluation methodologies that go beyond isolated model performance and instead assess detection behavior across multiple dimensions.

A further challenge arises from the absence of ground-truth labels. In unsupervised settings, traditional performance metrics such as precision and recall cannot be computed directly. Researchers must instead rely on indirect evaluation indicators, including score distributions, threshold stability, and inter-model agreement [3]. This methodological constraint underscores the importance of comparative and ensemble-based analyses to assess robustness and reduce model-specific biases in label-free environments.

Motivated by these challenges, this study conducts a systematic comparative analysis of unsupervised machine learning and deep learning techniques for anomaly detection in recommender systems. Using the MovieLens 1M dataset as a representative benchmark, we construct a user-level behavioral representation derived from statistical, temporal, and interaction-based features. We evaluate three widely adopted detection paradigms—Isolation Forest, One-Class Support Vector Machine, and an autoencoder-based neural network—and investigate their individual and collective behavior through an ensemble scoring strategy.

The contributions of this work are threefold. First, we provide a unified experimental framework for comparing classical machine learning and deep learning-based unsuper-

vised anomaly detection models under a consistent behavioral feature representation. Second, we introduce a comprehensive label-free evaluation protocol that combines score distribution analysis, percentile-based thresholding, ranking stability, and inter-model agreement. Third, we complement this evaluation with controlled experiments using synthetic attack profiles to systematically assess detection behavior across different adversarial strategies. Rather than proposing a novel detection algorithm, this study aims to provide a rigorous and reproducible comparative analysis that bridges methodological gaps in the evaluation of anomaly detection approaches for recommender systems.

The remainder of this paper is organized as follows. Section 2 reviews related work on recommender system security and anomaly detection. Section 3 describes the materials and methods, including the dataset, preprocessing strategy, feature engineering pipeline, and unsupervised detection models. Section 4 presents the experimental setup, including hyperparameter configuration and label-free evaluation protocols. Section 5 reports the experimental results, while Section 6 discusses their implications. Section 7 concludes the paper and outlines directions for future research.

2. Related Work

Research on security in recommender systems has primarily focused on the detection and mitigation of shilling or profile injection attacks, where malicious users attempt to manipulate recommendation outcomes by injecting biased ratings. Early foundational work by Lam and Riedl [2] demonstrated that collaborative filtering systems are vulnerable even to relatively small numbers of injected profiles. Subsequent studies expanded on attack modeling and defensive strategies, including feature-based classification approaches for identifying malicious users [3,4]. These works established the fundamental taxonomy of attack models and defensive mechanisms in collaborative recommender systems.

Several surveys have systematically analyzed vulnerabilities and countermeasures in recommender systems. Gunes et al. [7] provided a comprehensive overview of shilling attacks and detection strategies, highlighting the limitations of rule-based and heuristic methods. More recent studies have proposed increasingly sophisticated shilling attack strategies, including graph convolution-based generative models designed to bypass traditional detection mechanisms [8]. These developments illustrate the evolving nature of adversarial behavior and the difficulty of maintaining robust detection mechanisms in dynamic environments.

Traditional attack detection methods often rely on supervised learning frameworks, assuming the availability of labeled attack data [4]. However, in real-world deployments, explicit ground-truth labels identifying malicious users are rarely accessible. Moreover, legitimate users may exhibit behavioral patterns similar to attack profiles, further complicating classification. This limitation has motivated the exploration of anomaly detection approaches that treat malicious behavior as deviations from normal user activity rather than predefined classes.

Anomaly detection in recommender systems has been explored using statistical and machine learning methods. Yang et al. [9] proposed detecting abnormal user behavior through feature-based modeling of rating patterns. More broadly, unsupervised anomaly detection techniques have been extensively studied in the data mining literature [10,11], including tree-based methods such as Isolation Forest [12], kernel-based approaches such as One-Class Support Vector Machines [13], and neural network models such as autoencoders [14,15]. While these models have demonstrated effectiveness in general anomaly detection tasks, comparative evaluations within recommender system contexts remain limited.

Recent advances have also considered ensemble-based detection strategies to improve robustness. Ensemble approaches aim to aggregate heterogeneous anomaly signals and mitigate model-specific biases [16]. However, most existing studies either focus on synthetic attack injection scenarios or evaluate a single detection paradigm in isolation. Comparative analyses of classical machine learning and deep learning approaches under a unified behavioral representation remain relatively scarce.

In contrast to prior work that assumes labeled attacks or synthetic injection experiments, this study adopts a purely unsupervised perspective and evaluates multiple detection paradigms under a consistent feature extraction framework. By analyzing model agreement, threshold stability, and ensemble integration without relying on ground-truth labels, this work contributes to bridging the gap between theoretical attack modeling and practical anomaly detection in real-world recommender system settings.

While ensemble-based anomaly detection approaches have been explored in several machine learning contexts, many existing studies rely on supervised or semi-supervised settings where labeled attack profiles are available. In recommender system environments, however, such labeled data are rarely accessible due to the evolving nature of adversarial strategies and the difficulty of identifying malicious users with certainty.

The framework proposed in this study differs from prior ensemble-based approaches in three important aspects. First, it adopts a fully unsupervised perspective, integrating heterogeneous anomaly detection paradigms—including tree-based, kernel-based, and neural models—without requiring labeled attack data. Second, the proposed approach relies on a unified behavioral representation derived exclusively from rating interactions, combining statistical, temporal, and interaction-based signals that can be computed directly from standard recommender system logs. Third, the evaluation methodology emphasizes label-free validation indicators, including anomaly score distributions, percentile-based threshold stability, and cross-model agreement. Together, these elements provide a practical and interpretable framework for anomaly detection in recommender systems operating under realistic data constraints.

3. Materials and Methods

3.1. Dataset Description

The experiments conducted in this study are based on the MovieLens 1M dataset, a widely used benchmark in recommender systems research [17]. The dataset was released by the GroupLens Research Group and contains explicit user–item rating interactions collected from the MovieLens online platform.

MovieLens 1M consists of approximately one million ratings provided by 6040 users on 3706 movies, using a discrete rating scale from 1 to 5. Each rating is associated with a timestamp, enabling the analysis of temporal activity patterns in addition to rating distributions. The dataset also includes basic user metadata and item information; however, this study focuses exclusively on interaction data to ensure that the proposed detection framework relies only on observable behavioral signals commonly available in real-world recommender system deployments.

The MovieLens dataset is particularly suitable for the analysis of anomalous behavior due to its sparsity characteristics and heterogeneous user activity levels. While the dataset does not contain explicit labels identifying malicious or attack-driven users, it provides a realistic setting for unsupervised anomaly detection, where suspicious behavior must be inferred from deviations in rating, temporal, and interaction patterns rather than predefined attack annotations.

Prior to feature extraction, minimal filtering was applied to remove users and items with extremely low interaction counts, as such profiles do not provide sufficient information

for stable behavioral characterization. This step improves the reliability of statistical and temporal features without altering the intrinsic distributional properties of the dataset.

Overall, the MovieLens 1M dataset serves as a representative and reproducible benchmark for evaluating unsupervised detection models in recommender systems, enabling meaningful comparison with prior work while reflecting practical constraints encountered in real-world scenarios.

3.2. Preprocessing and Feature Engineering

3.2.1. Filtering Strategy

To ensure the robustness and stability of the extracted behavioral features, a minimal filtering strategy was applied to the MovieLens 1M dataset. Specifically, users and items with extremely low interaction counts were removed prior to feature extraction. Users with fewer than 20 ratings and items with fewer than 20 received ratings were excluded from the analysis.

This filtering step aims to mitigate the impact of highly sparse user profiles and rarely rated items, which may introduce noise and unreliable statistical estimates in user-level behavioral features. Very short interaction histories can lead to unstable estimates of rating distribution statistics, temporal activity patterns, and item popularity measures, thereby negatively affecting the performance and interpretability of detection models.

The chosen thresholds follow common practices in recommender systems research and anomaly detection studies, where a minimum level of interaction is required to ensure meaningful behavioral characterization. Importantly, this filtering does not alter the underlying rating distributions of the dataset, but rather improves the reliability of user representations used for subsequent detection tasks.

While this filtering strategy improves the statistical reliability of user-level features, it may introduce a selection bias by excluding low-activity users. In particular, certain types of adversarial behavior, such as low-volume or stealthy attacks operating in the long tail of the interaction distribution, may not be captured under this constraint. Therefore, the proposed framework is primarily designed to detect anomalies in sufficiently active user profiles, where behavioral patterns can be robustly characterized.

3.2.2. Feature Extraction Pipeline

Detection-Oriented Feature Engineering Pipeline. After preprocessing and filtering, a detection-oriented feature extraction pipeline was applied at the user level. Each user profile was represented as a fixed-length feature vector capturing statistical, temporal, and interaction-based characteristics derived exclusively from observed rating behavior.

First, rating-based statistical features were extracted, including the number of ratings, mean and standard deviation of ratings, minimum and maximum ratings, rating entropy, and the proportion of extreme ratings. These features aim to capture abnormal rating patterns, such as overly polarized behavior or low-diversity rating distributions, which are commonly associated with malicious or anomalous users.

Second, temporal activity features were computed using rating timestamps. These include inter-arrival time statistics, profile time span, rating frequency, and burstiness indicators. Temporal features are particularly relevant for detecting suspicious behavior characterized by unusually dense rating activity within short time intervals.

Third, item interaction features were derived based on item popularity statistics. For each user, the average, minimum, maximum, and variance of the popularity of interacted items were computed. These features capture tendencies toward interacting with highly popular or niche items, which may reflect manipulation or coordinated behavior.

All feature groups were subsequently merged into a unified user-level representation. Missing values were handled conservatively using zero imputation. This choice ensures consistency across feature vectors while avoiding the introduction of artificial statistical structure that could arise from mean or distribution-based imputation. In the context of anomaly detection, this approach preserves the original sparsity patterns of user behavior and prevents biasing reconstruction-based or distance-based models.

During the preparation of this manuscript, an AI-assisted language model (ChatGPT-5.3, OpenAI) was used solely for language refinement purposes, including improving grammar, clarity, and readability of the text. The tool was not used to generate scientific content, design the methodology, conduct experiments, or interpret the results. All scientific contributions, analyses, and conclusions presented in this work were developed and validated by the authors.

3.2.3. Rationale for Minimum Interaction Threshold

A minimum interaction threshold was applied to both users and items in order to ensure statistically reliable behavioral representations. Specifically, users and items with fewer than 20 ratings were removed prior to feature extraction. The use of minimum interaction thresholds is a common and well-established practice in recommender systems research. Previous studies have shown that very sparse user profiles lead to unstable estimates of rating distributions and temporal activity patterns, which can negatively impact downstream learning tasks, including classification and anomaly detection [18,19]. Koren et al. [18] demonstrated that collaborative filtering models and derived behavioral representations become increasingly unreliable when trained on extremely short user histories, as sparsity amplifies noise and biases model learning. Similarly, Ricci et al. [19] highlight that users with very limited interaction histories provide insufficient information to characterize preferences or behavioral patterns in a meaningful way. From an anomaly detection perspective, Chandola et al. [10] emphasize that reliable detection of abnormal behavior requires a sufficient number of observations per entity to establish a baseline of normal behavior. In the context of recommender systems, user profiles with very few ratings do not provide enough evidence to distinguish between genuine variability and anomalous activity. Empirically, several studies using the MovieLens dataset adopt minimum thresholds ranging from 10 to 20 interactions per user to balance data coverage and representation stability [20]. Following these established practices, a threshold of 20 ratings was selected as a conservative compromise that preserves the majority of users while ensuring robust feature estimation. Importantly, this filtering step does not modify the intrinsic characteristics of the dataset, but rather improves the reliability of user-level feature extraction by excluding profiles for which statistical and temporal features would be poorly defined.

3.3. Unsupervised Detection Models

Given the absence of ground-truth labels indicating malicious or attack-driven behavior in the MovieLens 1M dataset, this study adopts unsupervised learning approaches for attack detection. Unsupervised models are particularly suitable for this setting, as they aim to identify deviations from normal user behavior without requiring explicit annotations, which are rarely available in real-world recommender system datasets.

In this context, the detection task is formulated as a user-level anomaly detection problem, where each user is represented by a feature vector capturing rating statistics, temporal activity patterns, and item interaction characteristics. Users whose behavioral profiles significantly deviate from the majority are considered suspicious.

3.3.1. Isolation Forest

Isolation Forest was employed as a tree-based unsupervised anomaly detection method. Unlike distance-based techniques, Isolation Forest isolates anomalies by recursively partitioning the feature space using randomly selected features and split values. Anomalous instances are expected to be isolated more quickly than normal instances, resulting in shorter average path lengths [12].

This method is particularly well-suited for high-dimensional behavioral feature spaces and has been widely adopted in anomaly detection tasks due to its computational efficiency and robustness to irrelevant features. In the context of recommender systems, Isolation Forest enables the identification of users exhibiting unusual rating distributions or abnormal temporal activity without imposing assumptions about data distribution.

3.3.2. One-Class Support Vector Machine

One-Class Support Vector Machine (OC-SVM) was also considered as a classical unsupervised baseline. OC-SVM aims to learn a decision boundary that encloses the majority of normal data points in the feature space, treating observations outside this boundary as anomalies [13].

Although sensitive to feature scaling and kernel selection, OC-SVM remains a widely used benchmark in anomaly detection research. Its inclusion in this study provides a strong baseline for evaluating the effectiveness of tree-based and neural approaches under identical feature representations.

To formalize key behavioral features, rating entropy is defined as:

$$H(u) = - \sum_{r \in R} p_r \log(p_r)$$

where p_r represents the proportion of ratings with value r for user u , and R denotes the set of possible rating values.

Temporal burstiness is quantified using the proportion of interactions occurring within short time intervals:

$$B(u) = \frac{N_{\Delta t < \tau}}{N_{\text{total}}}$$

where $N_{\Delta t < \tau}$ is the number of consecutive interactions with inter-arrival time below a threshold τ , and N_{total} is the total number of interactions for user u .

3.3.3. Autoencoder-Based Anomaly Detection

To incorporate deep learning-based detection, an autoencoder model was employed. Autoencoders are neural networks trained to reconstruct their input data by learning a compressed latent representation. When trained primarily on normal behavioral patterns, the model achieves low reconstruction error for typical users, while anomalous users exhibit significantly higher reconstruction errors [14,15].

In this study, a fully connected autoencoder architecture was trained on standardized user-level features. The reconstruction error, measured as the mean squared error between the input and reconstructed output, was used as the anomaly score. This approach enables the model to capture non-linear relationships between behavioral features that may not be detected by classical methods.

3.3.4. Unified Detection Framework

All detection models were applied to the same standardized user-level feature representation to ensure methodological consistency and fair comparison. Anomaly scores were computed independently by each model under identical data preprocessing conditions.

3.4. Ensemble Scoring Strategy

Given that different unsupervised detection models capture distinct structural properties of user behavior, an ensemble scoring strategy was adopted to integrate heterogeneous anomaly signals into a unified ranking. Ensemble approaches are commonly used in anomaly detection to reduce model-specific bias and improve robustness by aggregating complementary detection perspectives [16].

In this study, anomaly scores produced by Isolation Forest, One-Class SVM, and the autoencoder were first normalized to a common scale using min–max normalization. This step ensures comparability across models whose raw score distributions may differ in magnitude and dispersion.

The ensemble anomaly score for each user was computed as the arithmetic mean of the normalized scores across models. Formally, let $s_i^{(m)}$ denote the normalized anomaly score assigned to user i by model $m \in \{IF, SVM, AE\}$. The ensemble score S_i is defined as:

$$S_i = \frac{1}{M} \sum_{m=1}^M s_i^{(m)},$$

where $M = 3$ corresponds to the number of detection models.

This simple aggregation strategy was intentionally selected to maintain interpretability and avoid introducing additional hyperparameters. More complex ensemble mechanisms, such as weighted averaging or stacking, were not considered in order to preserve the unsupervised and label-free nature of the framework.

By combining tree-based, kernel-based, and neural anomaly signals, the ensemble approach aims to identify users consistently exhibiting abnormal behavior across multiple detection paradigms, while mitigating the influence of model-specific sensitivities.

4. Experimental Setup

All experiments were conducted on the MovieLens 1M dataset using a user-level representation derived from explicit rating interactions. Following the preprocessing and filtering steps described in the previous section, each user was represented by a fixed-length feature vector capturing rating statistics, temporal activity patterns, and item popularity characteristics.

The experimental procedure followed the methodological framework described in the previous section. All detection models were applied to the standardized user-level feature representation derived from the MovieLens 1M dataset.

Three unsupervised detection models were evaluated: Isolation Forest, One-Class Support Vector Machine, and an autoencoder-based neural network. All models were trained on the same standardized feature set to ensure comparability across methods. Feature standardization was applied using z-score normalization, which is particularly important for distance- and kernel-based models such as One-Class SVM, as well as for neural network optimization [11].

4.1. Hyperparameter Configuration

Model hyperparameters were selected based on commonly adopted values in the anomaly detection literature. For Isolation Forest, the contamination parameter was used to control the expected fraction of anomalous users. Similarly, the parameter ν in One-Class SVM was configured to reflect the same expected anomaly rate, ensuring a fair alignment between methods [12,13].

The contamination parameter was aligned with the 2% percentile-based threshold to maintain consistency between model calibration and alerting strategy.

The autoencoder architecture consisted of fully connected layers with a symmetric encoder–decoder structure and was trained to minimize reconstruction error using mean squared error loss.

Table 1 summarizes the hyperparameter settings used for all detection models. To ensure reproducibility and avoid dataset-specific overfitting, commonly adopted values from the literature and default configurations were used without extensive tuning.

Table 1. Hyperparameter settings for the unsupervised detection models.

Model	Hyperparameter	Value
Isolation Forest	Number of trees ($n_{estimators}$)	300
	Subsample size ($max_samples$)	auto
	Contamination rate	0.02
	Parallel jobs (n_jobs)	−1
	Random seed	42
One-Class SVM	Kernel	RBF
	ν (upper bound on anomalies)	0.02
	γ	scale
	Shrinking	True
Autoencoder	Encoder layers	$[n_{in}, 64, 16]$
	Decoder layers	$[16, 64, n_{in}]$
	Activation function	ReLU
	Loss function	MSE
	Optimizer	Adam
	Learning rate	1×10^{-3}
	Training epochs/batch size	25/256

4.2. Synthetic Attack Injection

Although the MovieLens 1M dataset does not contain explicit labels identifying malicious users, controlled experiments with synthetic attack profiles were conducted to assess the detection capability of the proposed framework under different adversarial scenarios.

Following common practices in recommender system security research, three representative attack strategies were simulated: random, average, and bandwagon attacks. Each attack type generates synthetic user profiles that attempt to manipulate the recommendation process by rating a set of target items while interacting with additional filler items.

In the random attack, filler items are selected randomly and assigned ratings sampled uniformly from the available rating scale. This strategy produces user profiles that often deviate substantially from typical rating behavior.

The average attack generates filler ratings that approximate the average rating of each item in the dataset. By mimicking the global rating distribution, this attack produces profiles that resemble genuine user behavior more closely and are therefore more difficult to detect.

The bandwagon attack attempts to camouflage malicious activity by including ratings for a set of highly popular items in addition to the target items. This strategy exploits item popularity patterns to make injected profiles appear more consistent with the natural interaction structure of the system.

For each attack strategy, 80 synthetic user profiles were generated, resulting in a total of 240 injected attacker profiles. These synthetic users were merged with the original dataset prior to feature extraction, enabling the detection models to evaluate behavioral deviations under controlled conditions.

Importantly, these injected profiles were used exclusively for experimental validation and were not required for model training. The detection models remained fully unsupervised and were trained only on the observed user behavior without using attack labels.

The injected profiles enable controlled experiments for evaluating detection performance across different adversarial strategies while preserving the unsupervised training setup of the proposed framework.

4.3. Evaluation Protocol Without Ground Truth

Since no ground-truth labels are available, traditional supervised evaluation metrics such as accuracy, precision, and recall cannot be computed. Instead, the evaluation focuses on label-free indicators that are commonly employed in unsupervised anomaly detection settings.

First, score distribution analysis was conducted to examine the separation between typical users and high-scoring anomalous users. Heavy-tailed score distributions and clear upper-score regions provide evidence of model discrimination capability.

Second, agreement analysis across models was performed using Jaccard similarity over the top-K ranked users. High overlap between different detection methods suggests consistent identification of anomalous behavior, whereas low overlap may indicate model-specific biases [16].

Third, stability analysis was conducted by measuring the overlap of flagged users across different percentile thresholds (1%, 2%, 5%). Stable detection across thresholds supports the robustness of the anomaly scoring process.

Finally, a qualitative inspection of selected high-scoring users was performed by analyzing their original rating timelines and interaction patterns. This step provides interpretability and empirical validation, allowing the identification of behaviors such as rating bursts, extreme rating distributions, or unusually dense interaction profiles, which are often associated with suspicious activity in recommender systems.

4.4. Thresholding Strategy

Each detection model produces a continuous anomaly score for every user, where higher scores indicate a greater degree of deviation from typical behavior. To facilitate cross-model comparison, anomaly scores were normalized to a common scale using min-max normalization.

In the absence of labeled data, a thresholding strategy based on top-percentile selection was adopted. Specifically, users were ranked according to their anomaly scores, and the top $p \in \{1, 2, 5\}$. This approach is widely used in unsupervised anomaly detection and reflects practical alerting or inspection budgets in real-world systems [10,20].

Using multiple values of p enables the assessment of detection stability and sensitivity with respect to the assumed anomaly rate. Rather than claiming definitive identification of attacks, flagged users are interpreted as candidates exhibiting suspicious or anomalous behavioral patterns.

In practical deployments, evaluation metrics such as ROC-AUC and PR-AUC provide guidance on the discriminative capability of the detection models, while percentile-based thresholds translate these capabilities into actionable monitoring decisions. For example, higher ROC-AUC values indicate improved separation between typical and anomalous users, supporting the selection of stricter thresholds (e.g., top 1%), whereas lower separability may require broader thresholds to ensure adequate coverage of suspicious behavior.

Reproducibility Considerations

All preprocessing steps, feature extraction procedures, model configurations, and evaluation protocols were implemented using reproducible Jupyter notebooks. Intermediate

artifacts, including feature matrices and anomaly scores, were stored in CSV format to facilitate transparency and reuse. This design enables straightforward replication of the experiments and supports extensibility to additional datasets or detection models.

Although deep learning-based models such as autoencoders may exhibit variability across training runs due to random initialization and stochastic optimization, preliminary repeated experiments indicated consistent ranking patterns among the highest-scoring anomalous users. This suggests that, despite minor numerical variations, the overall detection behavior remains stable, particularly when integrated within the ensemble framework.

5. Results

5.1. Anomaly Score Distributions

The distribution of ensemble anomaly scores across users is illustrated in Figure 1. The score distribution exhibits a clear heavy-tailed pattern, where a relatively small subset of users receives substantially higher anomaly scores compared to the majority of the population.

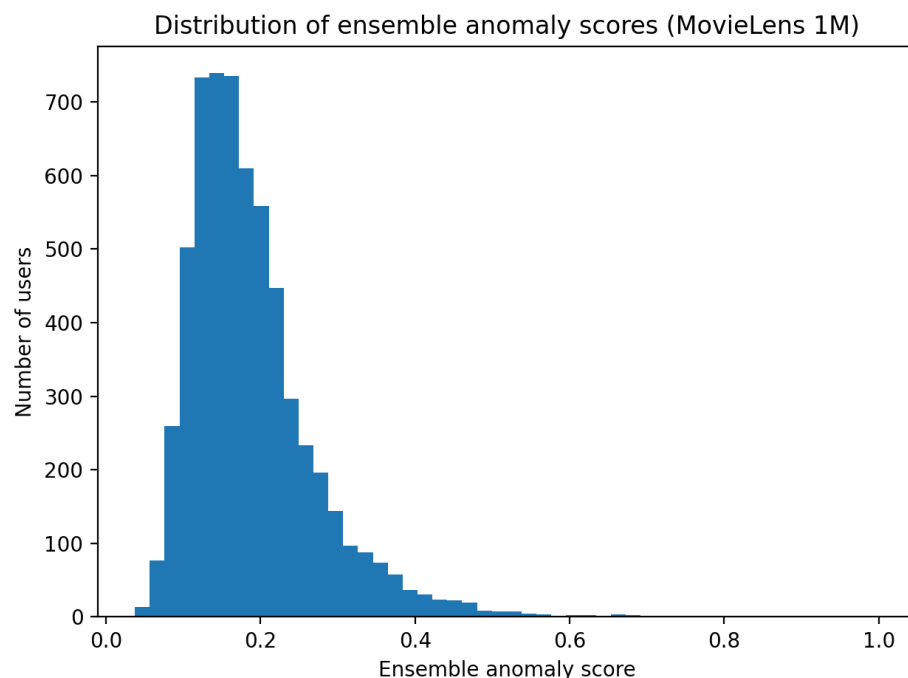


Figure 1. Distribution of ensemble anomaly scores across users in the MovieLens 1M dataset. The heavy-tailed distribution indicates that the majority of users exhibit typical behavioral patterns with low anomaly scores, while a small subset of users forms an upper tail with substantially higher scores. This separation suggests that the proposed framework is able to identify potentially deviant user profiles without relying on labeled attack data.

The concentration of users at lower anomaly score values is consistent with the expected behavior of unsupervised anomaly detection models, where most instances correspond to normal behavioral patterns. The presence of a distinct upper-score tail indicates that the proposed framework effectively differentiates highly deviant profiles without relying on labeled attack data.

Similar skewed distributions were observed for individual detection models, although differences in score dispersion were noticeable, motivating the subsequent integration of heterogeneous anomaly signals through ensemble scoring.

5.2. Threshold-Based Detection and Alerting Rates

To operationalize anomaly detection in the absence of ground-truth labels, a percentile-based thresholding strategy was adopted. Table 2 reports the anomaly score thresholds corresponding to the top 1%, 2%, and 5% of users ranked by the ensemble anomaly score.

Table 2. Percentile-based anomaly score thresholds using the ensemble model.

Top Percentile (%)	Threshold Value	Flagged Users
1	0.4617	61
2	0.4149	121
5	0.3477	302

As visualized in Figure 2, threshold values decrease smoothly as the alerting percentile increases, indicating stable ranking behavior across operating points.

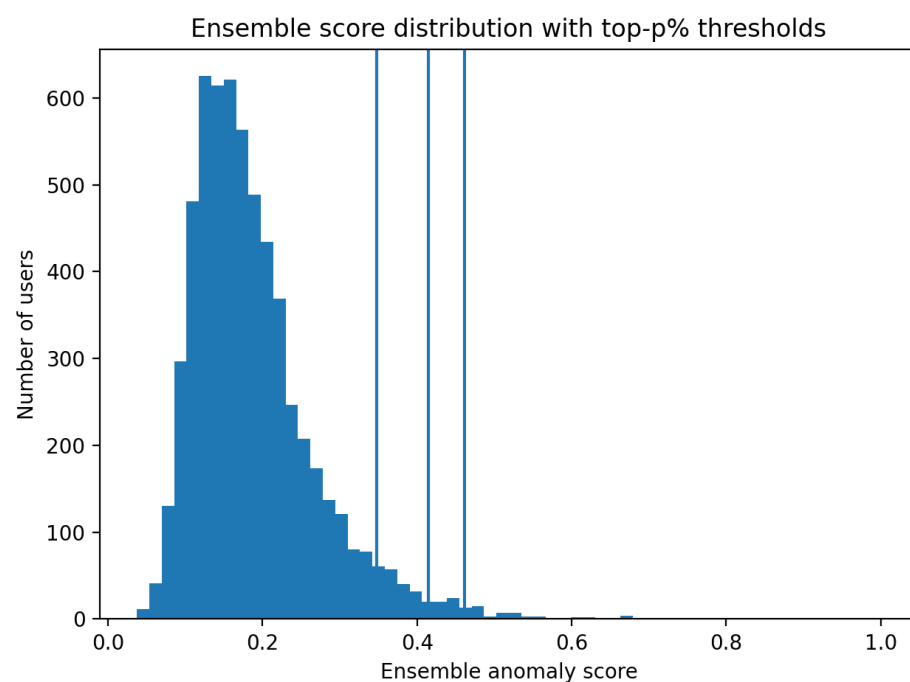


Figure 2. Ensemble anomaly score distribution with percentile-based thresholds corresponding to different alerting rates. The smooth progression of threshold values indicates stable ranking behavior across operating points, supporting the use of percentile-based thresholds as a practical mechanism for anomaly monitoring in label-free settings. Vertical lines indicate the corresponding threshold values used to flag anomalous users

The high concentration of users at lower anomaly score values does not indicate score saturation; rather, it reflects the model's ability to assign similar low scores to typical users while preserving a distinct tail for highly deviant profiles. Importantly, users identified at stricter thresholds are largely contained within broader thresholds, demonstrating coherent ranking stability.

5.3. Behavioral Characteristics of Flagged Users

To further characterize detected anomalies, aggregate feature analyses were conducted. Figures 3 and 4 illustrate the relationship between ensemble anomaly scores and representative behavioral indicators.

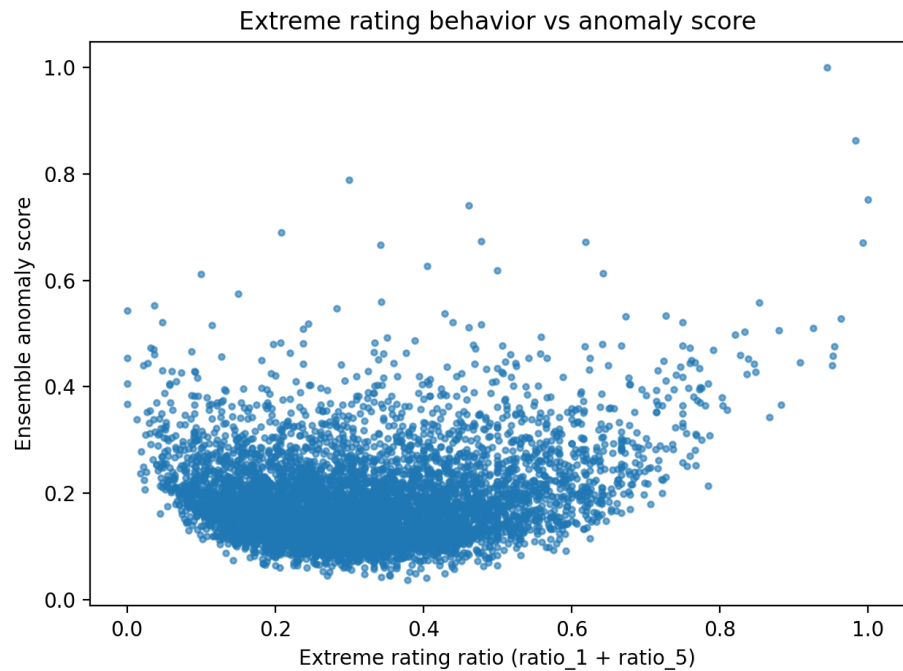


Figure 3. Relationship between extreme rating behavior and ensemble anomaly scores. Users with higher anomaly scores tend to exhibit a larger proportion of extreme ratings (e.g., minimum or maximum values), suggesting that unusual rating intensity may contribute to the identification of anomalous behavioral patterns.

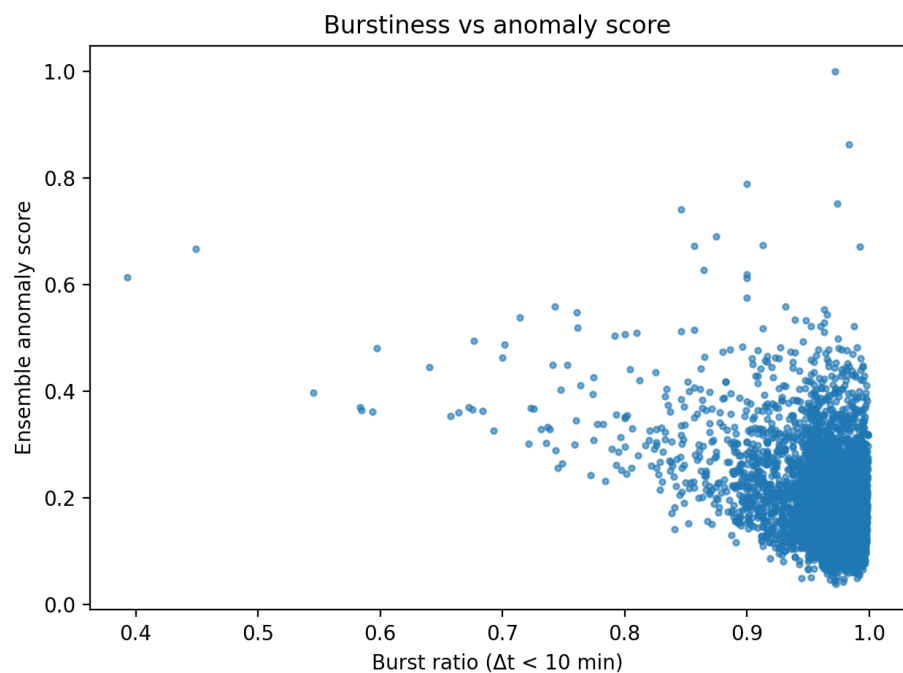


Figure 4. Relationship between temporal burstiness and ensemble anomaly scores. Profiles with highly concentrated rating activity within short time intervals tend to receive higher anomaly scores, indicating that temporal burst patterns are a relevant behavioral signal for detecting suspicious user activity.

Users with higher anomaly scores frequently exhibit elevated proportions of extreme ratings, unusually dense rating activity, or increased temporal burstiness. However, these behavioral signatures are not uniformly present across all flagged users. The observed vari-

ability indicates that anomalous behavior manifests across multiple behavioral dimensions rather than through a single dominant feature.

These exploratory observations motivate a more systematic analysis of which behavioral signals contribute most strongly to anomaly detection. The following sections therefore investigate feature importance and feature–score correlations to better interpret the behavioral drivers behind the anomaly scores produced by the framework.

5.4. Model Agreement Analysis

The consistency of detection results across models was evaluated using Jaccard similarity over the top- K ranked users. Table 3 reports pairwise agreement between detection methods.

Table 3. Agreement between unsupervised detection models measured by Jaccard similarity on the top- K ranked users.

Model Pair	Jaccard Similarity
Isolation Forest vs. One-Class SVM	0.282
Isolation Forest vs. Autoencoder	0.163
One-Class SVM vs. Autoencoder	0.282
Ensemble vs. Isolation Forest	0.471
Ensemble vs. One-Class SVM	0.449
Ensemble vs. Autoencoder	0.408

Agreement between individual models ranges from low to moderate, indicating that different detection paradigms capture partially distinct aspects of anomalous behavior. The ensemble exhibits consistently higher agreement with each individual model, suggesting improved stability in the integrated ranking.

While the previous analyses provide insight into the internal consistency and behavioral characteristics of the anomaly detection framework, they do not directly measure detection performance under controlled adversarial scenarios. To further evaluate the robustness of the proposed approach, additional experiments were conducted using synthetic attack profiles that simulate common manipulation strategies in recommender systems.

5.5. Ranking Stability Analysis

To further evaluate the consistency of the anomaly rankings produced by different detection models, additional ranking stability metrics were computed. While the Jaccard similarity presented in the previous subsection captures overlap among top- K detected users, it does not fully reflect the consistency of the ranking order.

To address this limitation, we evaluated ranking stability using Spearman’s rank correlation coefficient, which measures the monotonic agreement between full anomaly score rankings across models. This provides a more rigorous assessment of whether different models assign similar relative importance to users, beyond simple overlap at fixed thresholds.

Table 4 reports the Spearman correlation coefficients between anomaly score rankings produced by different models and the ensemble.

The results indicate a moderate to high correlation between the ensemble ranking and individual models, particularly for Isolation Forest and One-Class SVM. This suggests that the ensemble preserves the dominant ranking structure of the strongest individual models while reducing model-specific variability.

Table 4. Spearman rank correlation between anomaly score rankings across detection models.

Model Pair	Spearman ρ
Isolation Forest vs. Ensemble Mean	0.863
One-Class SVM vs. Ensemble Mean	0.778
Autoencoder vs. Ensemble Mean	0.662
Isolation Forest vs. One-Class SVM	0.726
Isolation Forest vs. Autoencoder	0.601
One-Class SVM vs. Autoencoder	0.589

Lower correlations involving the autoencoder reflect its sensitivity to different aspects of the feature space, highlighting the complementary nature of neural-based detection compared to classical methods. Overall, the ensemble approach achieves a balanced integration of heterogeneous anomaly signals while maintaining a stable and consistent ranking of suspicious users.

5.6. Detection Performance Across Attack Types

To complement the label-free evaluation presented above, controlled experiments using the synthetic attack profiles described in Section 4.2 were conducted.

These experiments aim to evaluate the ability of the proposed framework to identify anomalous user behavior under different adversarial strategies.

Three commonly studied attack strategies in recommender systems were simulated: random, average, and bandwagon attacks.

In the random attack, filler ratings are assigned randomly across items, producing profiles that deviate strongly from typical user behavior. The average attack generates ratings that approximate the average rating of each item, making malicious profiles resemble normal users more closely. In the bandwagon attack, injected users rate a set of highly popular items in addition to target items in order to camouflage their behavior within the natural popularity structure of the system.

For each attack strategy, a one-vs-rest evaluation was performed in which injected users were treated as the positive class and all remaining users were treated as negatives. Detection performance was evaluated using ROC-AUC and PR-AUC metrics.

Table 5 summarizes the ROC-AUC scores obtained for each attack type across the evaluated detection models. The results reveal that detection difficulty varies substantially depending on the attack strategy.

Random attacks are the easiest to detect, with Isolation Forest achieving the highest ROC-AUC score (0.936). This behavior is expected since random attacks produce rating patterns that strongly deviate from the behavioral distributions observed in genuine users.

Bandwagon attacks exhibit intermediate detection difficulty. Although attackers attempt to camouflage their profiles by interacting with popular items, deviations in temporal activity patterns and interaction statistics still allow anomaly detection models to identify suspicious users. In this scenario, the One-Class SVM achieves the highest ROC-AUC score (0.903).

In contrast, average attacks are the most challenging to detect. These attacks deliberately mimic the global rating distribution of the dataset, making injected profiles more similar to genuine users. Nevertheless, the behavioral features used in the proposed framework still enable meaningful discrimination, with Isolation Forest achieving a ROC-AUC score of 0.876.

These results highlight that the detectability of malicious behavior depends not only on the anomaly detection algorithm but also on the behavioral characteristics of the attack strategy. Attacks that produce strong deviations from normal interaction patterns, such

as random attacks, are detected more easily. In contrast, attacks designed to mimic the statistical properties of genuine user behavior, such as average attacks, present greater detection difficulty.

Overall, the results demonstrate that the proposed behavioral feature framework is capable of identifying anomalous interaction patterns across multiple adversarial strategies. This controlled evaluation complements the label-free analyses presented earlier and provides additional empirical evidence supporting the robustness of the proposed anomaly detection approach.

Table 5. ROC-AUC performance across different attack types.

Model	Random	Bandwagon	Average
Isolation Forest	0.936	0.753	0.876
Ensemble Mean	0.927	0.863	0.860
Ensemble Weighted (Oracle)	0.929	0.860	0.864
Ensemble Max	0.922	0.842	0.844
One-Class SVM	0.877	0.903	0.794
Autoencoder	0.658	0.859	0.452

It is important to note that the experimental setup assumes a fixed proportion of injected attack profiles relative to genuine users. While this controlled ratio facilitates consistent comparison across attack types, detection performance may vary under different adversarial densities. In particular, higher attack injection rates may increase detectability due to stronger deviations in global behavioral distributions, whereas lower ratios may result in more subtle anomalies that are harder to distinguish from natural user variability. Nevertheless, the relative performance trends observed across models are expected to remain consistent, as they primarily reflect the structural characteristics of each attack strategy rather than the absolute proportion of adversarial profiles.

5.7. Feature Importance Analysis

To better understand which behavioral signals contribute most to anomaly detection, an additional feature importance analysis was conducted. Since the primary detection framework operates in a fully unsupervised setting, feature importance cannot be directly derived from the anomaly detection models themselves.

To address this limitation, a supervised proxy analysis was performed using the synthetic attack labels introduced in Section 4.2. A Random Forest classifier was trained on the same user-level feature representation to estimate feature importance using Gini importance. This proxy analysis does not alter the unsupervised detection framework but provides insight into which behavioral signals are most relevant for distinguishing between genuine and injected attack profiles.

Figure 5 presents the top ten most influential features identified by the proxy model. The results indicate that interaction intensity and temporal activity features play a central role in detecting anomalous user behavior. In particular, the number of ratings and the overall profile time span emerge as the most influential signals, suggesting that abnormal interaction density and unusually short or concentrated activity periods are key indicators of suspicious behavior.

Item popularity statistics also appear among the most important signals. Features such as the mean and standard deviation of item popularity capture whether users interact disproportionately with highly popular or unusually obscure items. Such patterns are commonly associated with manipulation strategies in recommender systems.

Temporal burstiness indicators, such as the proportion of ratings occurring within short time intervals, also contribute significantly to detection. These findings reinforce the

importance of incorporating temporal behavioral features when identifying anomalous interaction patterns in recommender system environments.

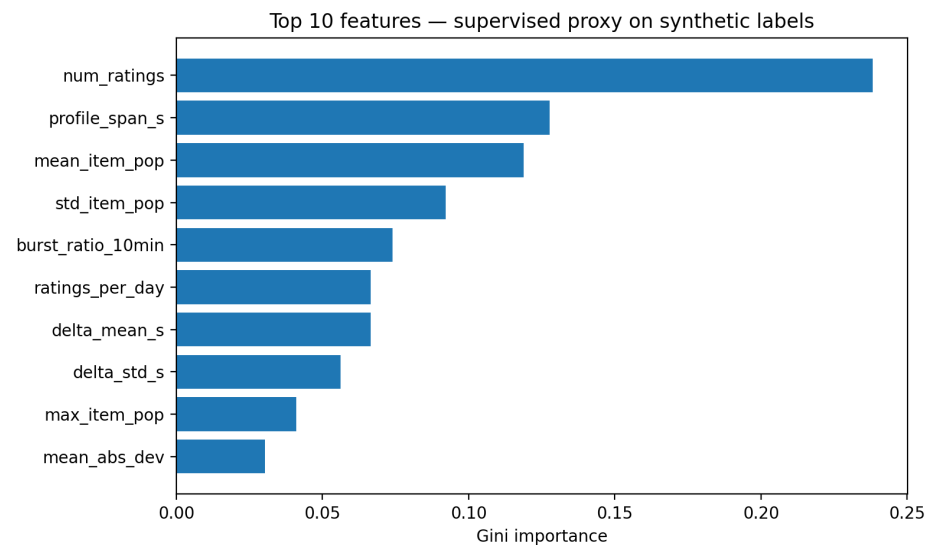


Figure 5. Top behavioral features ranked by Gini importance using a supervised proxy model trained on synthetic attack labels. Interaction intensity, temporal activity patterns, and item popularity statistics emerge as the most influential signals for distinguishing anomalous user profiles.

5.8. Feature–Score Correlation Analysis

To further interpret the behavior of the anomaly scoring mechanism, correlations between individual behavioral features and the ensemble anomaly score were analyzed using Spearman’s rank correlation coefficient. This analysis provides insight into which behavioral signals are most strongly associated with elevated anomaly scores.

To complement the visual analysis presented in Figure 6, Table 6 reports the behavioral features with the strongest Spearman correlations with the ensemble anomaly score.

Table 6. Top behavioral features most strongly correlated with the ensemble anomaly score (Spearman correlation).

Feature	Spearman ρ
burst_ratio_10 min	−0.262
delta_mean_s	0.192
ratings_per_day	−0.189
mean_item_pop	−0.181
min_item_pop	−0.172
profile_span_s	0.154
std_rating	−0.146
rating_entropy	−0.139
num_ratings	−0.133
burst_ratio_1 h	−0.127

The results confirm that temporal burstiness indicators and interaction frequency features are among the strongest correlates of anomalous behavior. In particular, short-term burst activity (e.g., burst_ratio_10min) shows the strongest correlation with anomaly scores, suggesting that highly concentrated rating activity is a key signal of suspicious behavior. Additionally, item popularity statistics and interaction intensity measures also exhibit notable correlations, indicating that anomalous users tend to interact with items in patterns that differ from typical user behavior.

Figure 6 illustrates the ten features with the strongest correlations with the ensemble anomaly score. The results reveal that temporal burstiness indicators exhibit the strongest relationship with anomaly scores. In particular, the proportion of ratings occurring within short time windows shows a negative correlation with the ensemble score, suggesting that users with highly concentrated activity bursts are more likely to be flagged as anomalous.

Interaction frequency features, such as ratings per day and inter-arrival time statistics, also demonstrate notable correlations with anomaly scores. These signals reflect deviations in user activity intensity that may indicate suspicious behavior. Additionally, item popularity statistics show moderate correlations, suggesting that anomalous users often interact with items whose popularity patterns differ from those of typical users.

Overall, these findings indicate that the ensemble anomaly score captures a combination of temporal, statistical, and interaction-based deviations, reinforcing the effectiveness of the proposed behavioral representation for detecting abnormal user activity.

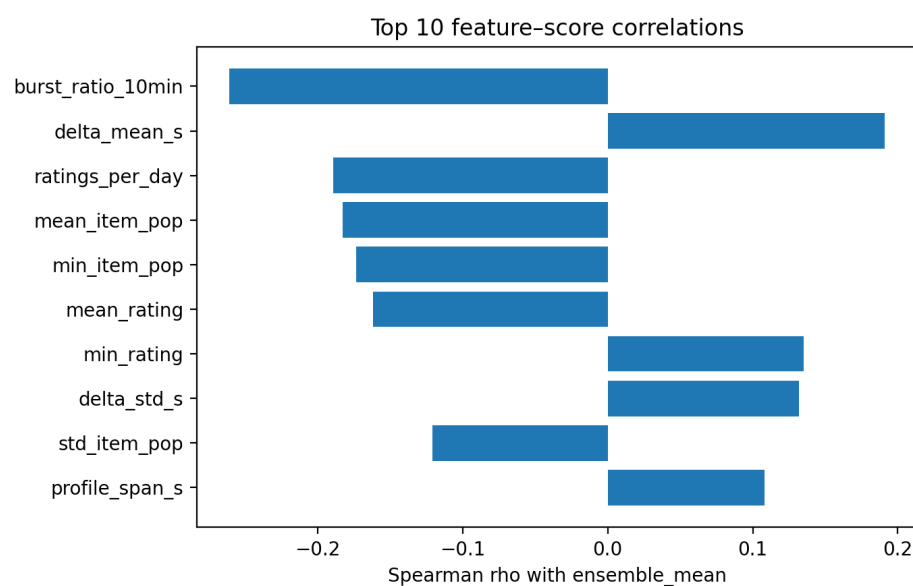


Figure 6. Top behavioral features ranked by Spearman correlation with the ensemble anomaly score. Temporal burstiness and interaction frequency features show the strongest relationships with anomalous behavior.

5.9. Comparison of Ensemble Aggregation Strategies

Because the proposed framework integrates heterogeneous anomaly detection models, the choice of score aggregation strategy may influence the stability and interpretability of the final anomaly ranking. To justify the use of mean aggregation in the proposed framework, additional experiments were conducted comparing alternative score aggregation methods. In addition to the mean aggregation used in the proposed framework, two alternative approaches were evaluated: maximum score aggregation and a weighted ensemble variant. To further illustrate the differences between aggregation strategies, Figure 7 presents the distribution of anomaly scores obtained using mean and max ensemble aggregation on the original dataset. The mean aggregation produces a smoother score distribution with a clearer separation between typical users and the upper anomaly tail. In contrast, the max aggregation strategy yields a wider dispersion of anomaly scores, which may introduce greater sensitivity to extreme model outputs. This behavior supports the use of mean aggregation as a more stable and robust mechanism for integrating heterogeneous anomaly signals.

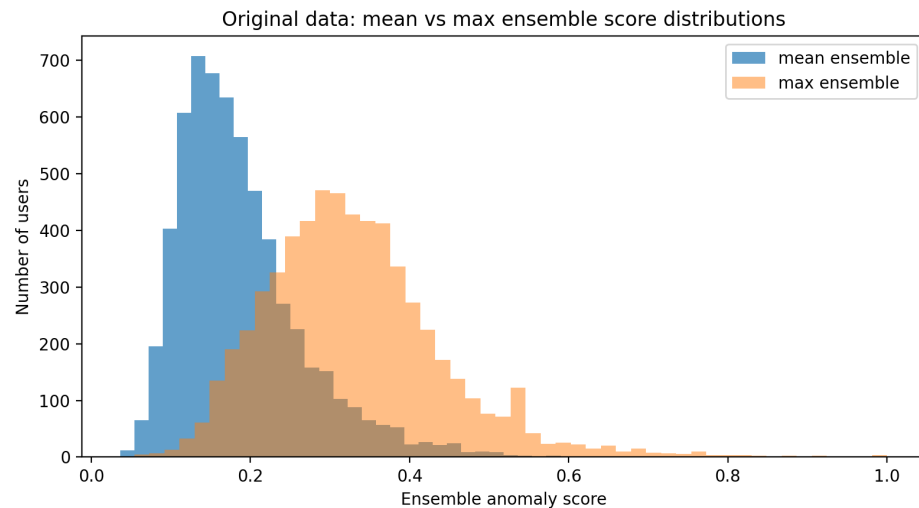


Figure 7. Distribution of anomaly scores produced by mean and max ensemble aggregation strategies on the original dataset. Mean aggregation produces a smoother score distribution and a more stable anomaly ranking compared to max aggregation.

Figure 8 summarizes the comparative detection performance across models using synthetic attack experiments. The results indicate that mean aggregation achieves performance comparable to the weighted ensemble variant while maintaining superior interpretability and avoiding additional hyperparameter tuning. In contrast, maximum aggregation exhibits slightly lower detection performance, suggesting that relying on the most extreme model score may introduce instability in the anomaly ranking.

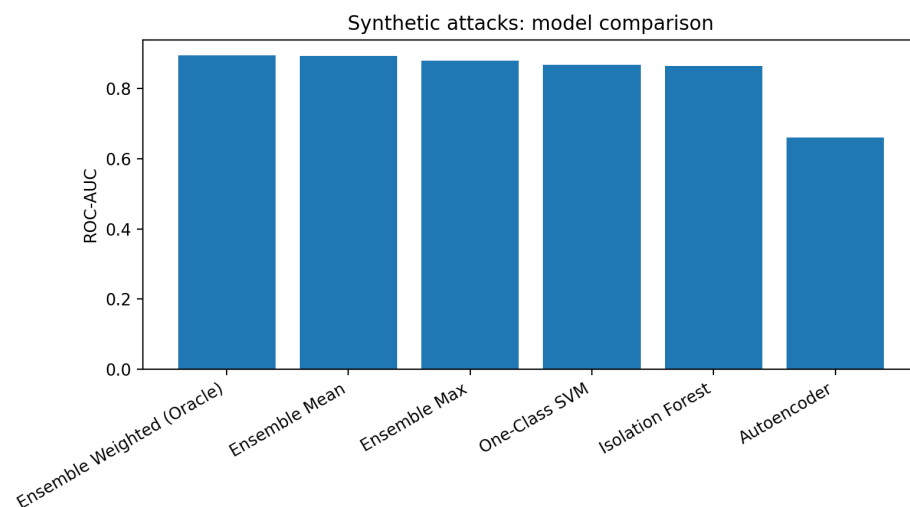


Figure 8. Comparison of detection performance across models using synthetic attack experiments. Ensemble aggregation achieves strong and stable performance across different detection paradigms.

These findings support the use of simple arithmetic averaging as a robust and interpretable ensemble strategy for integrating heterogeneous anomaly signals. By aggregating tree-based, kernel-based, and neural anomaly scores, the ensemble approach reduces model-specific sensitivities while preserving stable detection performance across attack scenarios.

Overall, the experimental results demonstrate that the proposed behavioral representation, combined with ensemble-based unsupervised detection, provides a robust framework for identifying anomalous user behavior in recommender systems across multiple evaluation perspectives.

While mean aggregation improves stability and robustness, it may also attenuate strong anomaly signals produced by individual models that are particularly sensitive to specific attack patterns. This trade-off reflects a balance between robustness and sensitivity. In practice, mean aggregation prioritizes consistent cross-model agreement over isolated extreme detections, which aligns with the objective of identifying users exhibiting broadly anomalous behavior across multiple perspectives.

6. Discussion

6.1. Heterogeneity of Anomalous Behavior

The results demonstrate that anomalous user behavior in recommender systems is inherently heterogeneous. Flagged users do not form a homogeneous group characterized by a single statistical property; instead, deviations emerge across multiple behavioral dimensions, including rating extremity, temporal density, and interaction patterns.

This heterogeneity explains the limited pairwise agreement observed between individual detection models. Tree-based, kernel-based, and neural approaches emphasize different structural properties of the feature space, leading to complementary anomaly rankings.

It is important to emphasize that the primary contribution of this work lies not in proposing a new anomaly detection algorithm, but in establishing a systematic and reproducible evaluation framework for analyzing unsupervised detection methods in recommender systems. By integrating heterogeneous models, behavioral feature engineering, and both label-free and controlled evaluation strategies, the study provides a structured approach for understanding detection behavior under realistic constraints.

6.2. Effectiveness of Ensemble Integration

The ensemble scoring strategy mitigates model-specific sensitivities by aggregating heterogeneous anomaly signals. The observed increase in agreement between the ensemble and individual models indicates that aggregation enhances ranking stability without enforcing strict consensus.

The use of simple arithmetic averaging preserves interpretability while avoiding additional hyperparameter tuning. This design choice is particularly relevant in unsupervised settings, where labeled validation data are unavailable.

6.3. Practical Implications for Recommender System Monitoring

From a practical standpoint, the percentile-based thresholding strategy aligns with realistic operational scenarios in which monitoring capacity or manual inspection budgets are limited. By allowing flexible control over the proportion of flagged users, the framework can be adapted to different deployment constraints.

Although the absence of labeled attack data prevents definitive validation of malicious behavior, the consistent identification of statistically deviant user profiles suggests that unsupervised ensemble-based detection can serve as an effective first-layer screening mechanism in recommender systems.

6.4. Limitations

Several limitations of this study should be acknowledged.

First, the analysis was conducted using a single publicly available dataset with explicit feedback. Although MovieLens 1M provides a widely accepted benchmark in recommender systems research, the absence of labeled attack data prevents definitive validation of detected anomalies as malicious or adversarial behavior. The proposed framework identifies statistically deviant user profiles rather than confirmed attacks.

Second, the detection approach operates at the user level and does not explicitly model relational structures such as user–user similarity networks or item co-rating graphs. Consequently, coordinated or group-based manipulation strategies may not be fully captured by the current feature representation.

Third, the ensemble scoring strategy employs simple arithmetic averaging without adaptive weighting. While this choice preserves interpretability and avoids additional hyperparameter tuning, more sophisticated aggregation strategies could potentially improve detection sensitivity under certain conditions.

Finally, the evaluation protocol relies on label-free indicators such as score distribution analysis and model agreement metrics. Although these methods are appropriate for unsupervised settings, they cannot provide quantitative measures of detection accuracy in the absence of ground-truth annotations.

To partially address this limitation, controlled experiments with synthetic attack profiles were conducted, as described in Section 4.2. These experiments enable a comparative evaluation of detection performance across different adversarial strategies, including random, average, and bandwagon attacks. While such synthetic scenarios cannot fully reproduce real-world adversarial behavior, they provide a controlled benchmark for assessing the sensitivity of the proposed detection framework under known attack patterns.

These limitations suggest that the proposed framework should be interpreted as a robust anomaly screening mechanism rather than a definitive attack detection solution.

7. Conclusions

This study proposed an unsupervised framework for detecting anomalous user behavior in recommender systems using behavioral features derived exclusively from explicit rating interactions. By combining statistical, temporal, and interaction-based representations, the framework transforms raw user–item data into a structured feature space suitable for label-free anomaly detection.

The comparative evaluation of Isolation Forest, One-Class Support Vector Machine, and an autoencoder-based model demonstrated that no single detection paradigm consistently captures all forms of behavioral deviation. Instead, the findings highlight the inherently heterogeneous nature of anomalous behavior in recommender systems, where different detection models emphasize complementary structural properties of user activity. The ensemble-based scoring strategy effectively integrates these heterogeneous signals, resulting in a more stable and consistent ranking of suspicious users.

The adoption of percentile-based thresholding further enables practical deployment by aligning anomaly detection with realistic inspection capacities. The observed stability of anomaly rankings across multiple threshold levels supports the robustness of the proposed approach under label-free conditions.

Additional controlled experiments with synthetic attack profiles further confirm that the proposed framework can detect anomalous behavior across multiple adversarial strategies, reinforcing its robustness for practical recommender system monitoring.

Overall, the results suggest that ensemble-based unsupervised detection constitutes a viable and interpretable first-layer screening mechanism for monitoring user behavior in recommender systems.

Future Research Directions

Future research may extend this framework in several directions. First, evaluating the approach on additional datasets, including implicit-feedback environments, would strengthen external validity. Second, integrating graph-based representations could enable detection of coordinated or group-based manipulation strategies that are not explic-

itly modeled in the current user-level formulation. Third, semi-supervised and weakly supervised approaches may offer improved sensitivity when limited labeled examples become available.

Finally, incorporating temporal evolution modeling may enhance the capacity to detect emerging anomalous patterns and support early-warning mechanisms in dynamic recommender system deployments.

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